Mohamed Wafik

Department of Mathematics, University of South Carolina
1523 Greene Street, LeConte College, Columbia, SC 29208, USA
mwafik@sc.edu · m-wafik.github.io · ORCID 0000-0001-7336-6570
Publications appear under Mohamed Wafik. Legal name: Mohamed Wafik ElSheikh.

RESEARCH INTERESTS

Arithmetic Statistics; Arithmetic Dynamics; Analytic and Algebraic Number Theory.

EDUCATION

Ph.D. in Mathematics, University of South Carolina, Columbia, SC, USA *2022 - Present*
Area: Arithmetic Statistics. Advisor: Frank Thorne. GPA: 4.0/4.0.

M.Sc. in Mathematics, Sabancı University, Istanbul, Turkey *2020 - 2022*
Thesis: "Primitive Prime Divisors in the Critical Orbit of Polynomial Dynamical Systems." Advisor: Mohammad Sadek. GPA: 4.0/4.0.

B.S. in Electronics and Communication Engineering, American University in Cairo, Egypt *2013 - 2018*
Minors: Computer Science, Mathematics. GPA: 3.83/4.0.

PUBLICATIONS

1. M. Sadek, M. Wafik, and T. Yesin, "Arithmetic progressions in polynomial orbits," *International Journal of Number Theory*, 20(8) (2024), 2009-2025. arXiv:2403.04397. doi:10.1142/S1793042124500970.
2. M. Sadek and M. Wafik, "Arbitrarily many primitive prime divisors in an iterate of $x^d + c$," submitted, arXiv:2209.07624.
3. M. Wafik, "Finite Fourier transform of doubly singular binary quintic forms," preprint.

RESEARCH EXPERIENCE

Doctoral Research, University of South Carolina. Advisor: Frank Thorne *2022 - Present*

- Compute finite Fourier transforms of invariant functions on spaces of binary forms, extending the work of Bhargava, Taniguchi and Thorne on spaces of lower degree binary forms to the setting of binary quintic forms.
- Reduce the computation to point counts on auxiliary varieties indexed by splitting type, the most delicate governed by a hyperelliptic curve arising from the Hessian covariant.

M.Sc. Thesis Research, Sabancı University. Advisor: Mohammad Sadek *2020 - 2022*

- Established construction results for primitive prime divisors in the critical orbit of $x^d + c$ over $\mathbb{Q}$ and local fields; developed proofs with computational evidence gathered with Magma and Mathematica.
- Results extended into two subsequent papers on divisibility conditions on critical orbits and coverings of polynomial orbits by arithmetic progressions.

TEACHING EXPERIENCE

Instructor of Record, Calculus Workshop II, University of South Carolina — *Fall 2026*

Instructor of Record, Discrete Math for Computer Science, University of South Carolina — *Aug 2025 - May 2026*

Instructor of Record, Calculus for Business and Social Science, University of South Carolina — *Jan 2025 - May 2025*

Instructor of Record, Intensive Basic College Math, University of South Carolina — *Aug 2024 - Dec 2024*

Instructor of Record, Calculus for Business and Social Science, University of South Carolina — *Aug 2023 - Jul 2024*

Teaching Assistant, Calculus 2, University of South Carolina — *Jan 2023 - May 2023*

Teaching Assistant, Calculus 1, University of South Carolina — *Aug 2022 - Dec 2022*

Teaching Assistant, Discrete Mathematics, Sabancı University — *Feb 2022 - Jun 2022*

Teaching Assistant, Linear Algebra, Sabancı University — *Sep 2021 - Jan 2022*

Teaching Assistant, Calculus 2, Sabancı University — *Feb 2021 - Jun 2021*

Teaching Assistant, Calculus 1, Sabancı University — *Sep 2020 - Jan 2021*

Undergraduate Teaching Assistant, Calculus IV, American University in Cairo — *Sep 2016 - Jan 2017, Sep 2017 - Jan 2018*

Undergraduate Teaching Assistant, Mechanics II (Dynamics), American University in Cairo — *Feb 2015 - Jun 2015, Feb 2016 - Jun 2016*

Undergraduate Teaching Assistant, Linear Algebra, American University in Cairo — *Feb 2018 - Jun 2018*

Undergraduate Teaching Assistant, Calculus II, American University in Cairo — *Feb 2014 - Jun 2014*

GRANTS, FELLOWSHIPS, AND AWARDS

Outstanding First Year Graduate Student, University of South Carolina — *2022 - 2023*

TÜBİTAK Scholarship (Scientific and Technological Research Council of Turkey), Sabancı University — *2021 - 2022*

Full Tuition Scholarship, Sabancı University — *2020 - 2021*

Highest Honors Achievement, Office of the Registrar, American University in Cairo — *2018*

Academic Achievement Scholarship, Office of Student Financial Aid, American University in Cairo — *2013 - 2018*

Honors Achievement, School of Sciences and Engineering, American University in Cairo — *2016, 2017, 2018*

TALKS

Palmetto Number Theory Series (PANTS XLI), Georgia Tech, Atlanta, GA, USA — *Oct 2026*
"Finite Fourier transform of doubly singular binary quintic forms."

FOUVRY73 Summer School, Bernoulli Center, EPFL, Lausanne, Switzerland, short talk — *Aug 2026*
"Finite Fourier transform of doubly singular binary quintic forms."

South Eastern Regional Meeting on Numbers (SERMON), High Point University, High Point, NC, USA *Apr 2026*
"Exponential sums and the Fourier transform of binary forms over finite fields."

32nd Journées Arithmétiques, University of Lorraine, Nancy, France, short talk *Jul 2023*
"Construction of polynomials with prescribed divisibility conditions on the critical orbit."

Palmetto Number Theory Series (PANTS XXXV), University of South Carolina, Columbia, SC, USA *Dec 2022*
"Construction of polynomials with prescribed divisibility conditions on the critical orbit."

Bahar Matematik Buluşmaları (Spring Mathematics Meetings), Turkey, invited talk *Oct 2021*
"Primitive prime divisors of integer sequences."

CONFERENCES AND WORKSHOPS

Algorithmic Number Theory Symposium (ANTS XVII), Bernoulli Institute, Groningen, Netherlands *Jul 2026*

SMS 2026: Universal Statistics in Number Theory, Université de Montréal, Montréal, Québec, Canada *May 2026*

Palmetto Number Theory Series (PANTS XXXVI), Clemson University, Clemson, SC, USA *Oct 2023*

Arizona Winter School 2023: Unlikely Intersections, University of Arizona, Tucson, AZ, USA *Mar 2023*

SuSAAN Summer School, Nesin Mathematics Village, İzmir, Turkey *Jun 2022*

CIMPA Preparation Course, Boğaziçi University, Istanbul, Turkey *Aug 2021*

NON-ACADEMIC EXPERIENCE

Junior Validation Engineer, Valeo, Cairo, Egypt *Dec 2018 - Sep 2020*

- Automated test-case execution for park-maneuver-assist functionality with Python; reviewed requirements for testability.

Data Science Intern, Teradata, Cairo, Egypt *Aug 2017*

Circuit Design Intern, Indian Mesh Pvt. Ltd., Chandigarh, India *Aug 2016*

EXTRACURRICULAR ACTIVITIES

Vice Chairman, IEEE Student Branch, American University in Cairo *2015 - 2018*

OGCDP Member, AIESEC, American University in Cairo *2014 - 2015*

SKILLS

Technical: C++, Java, Python, Mathematica, Magma, LaTeX (proficient); SQL, MATLAB, Cadence, VHDL, CAPL (intermediate).

Languages: English and Arabic (fluent); Turkish (intermediate).

REFERENCES

Available upon request.